Testing of hypothesis

Hypothesis: A base statement regarding Population Parameter which is to be tested is called hypothesis

Types:

① **Null hypothesis:-** A hypothesis which is unbiased and having no difference is called null hypothesis. It is denoted by H_0.

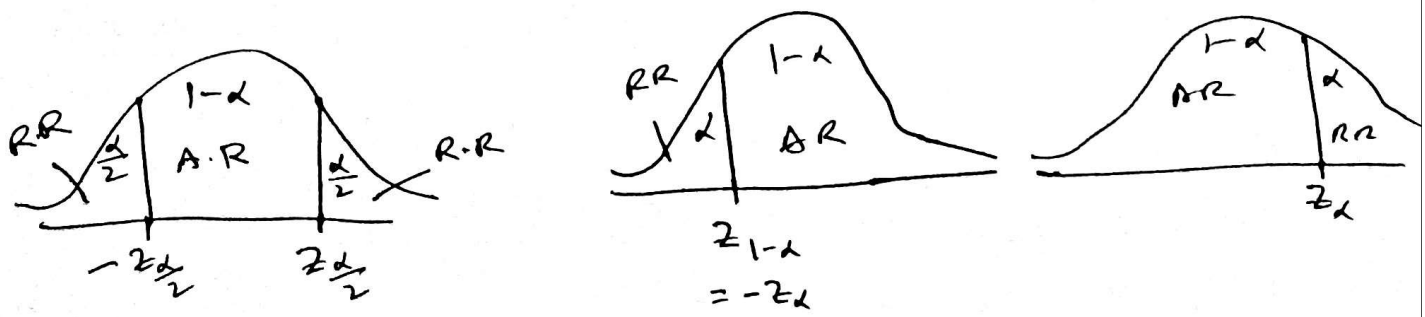
② **Alternative hypothesis:** A hypothesis which contradicts null hypothesis is called alternative hypothesis. It is denoted by H_1.

Examples:

① $H_0: \mu = 65$ kg
$H_1: \mu \neq 65$ kg
(Two tailed)

② $H_0: \mu \geq 8$
$H_1: \mu < 8$
(Left tail)

③ $H_0: \mu \leq 200$
$H_1: \mu > 200$
(Right tail)



Errors in Sampling:

1. Type-I error: Rejection of H_0 even when it is true.

i.e Null hypothesis H_0 is true but it is rejected by the test Procedure.

Probability (type I error) $= \alpha$

where 'α' is called level of Significance.

→

2. Type-II error:- Accepting H_0 even when it is false

i.e Null hypothesis H_0 is false but it is accepted by the test Procedure

Probability (type-II error) $= \beta$

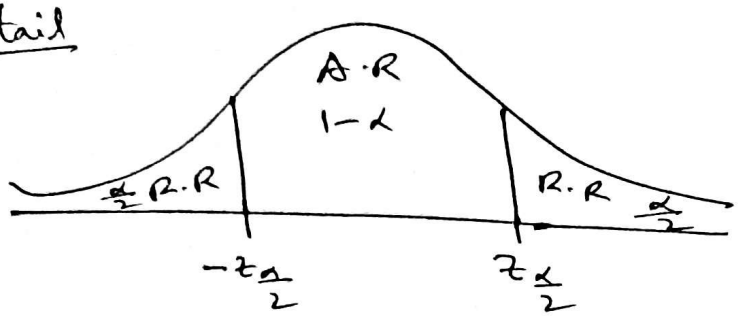
→ $1-\beta$ is called Power of the test.

Critical region: A region Corresponding to test-statistic in the Sample Space which lead to rejection of null hypothesis H_0 is called critical region (or) rejection region

→ The region in which the null hypothesis H_0 is accepted is called acceptance region.

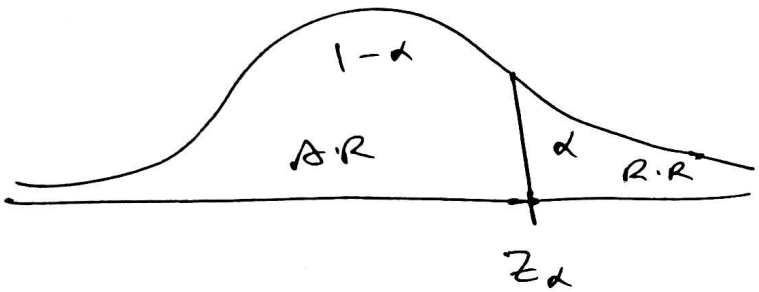
Critical values: (Significant values): The value (values) of test statistic which divide the total Sample Space into acceptance region and rejection region are called critical values or Significant values.

(i) <u>Two tail</u>



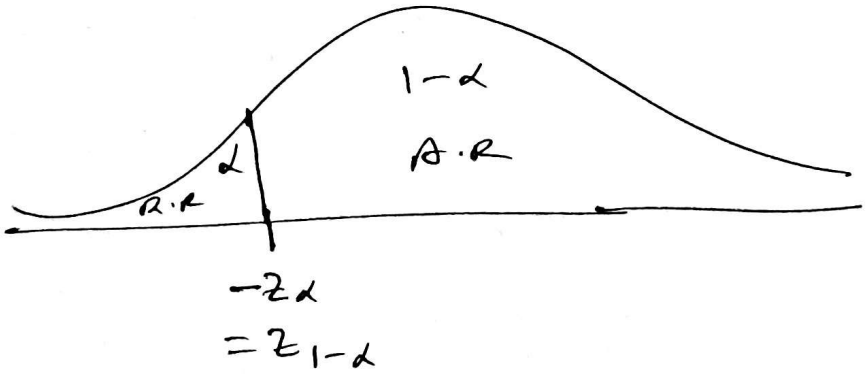
Critical values are $z_{\frac{\alpha}{2}}$ and $-z_{\frac{\alpha}{2}}$.

(ii) <u>Right tail:</u>



Critical value is z_α

(iii) <u>Left tail</u>



Critical value is $-z_\alpha$ & $z_{1-\alpha}$.